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Bayesian nonparametric model for heterogeneous treatment effects with zero-inflated data

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Abstract

One goal of precision medicine is to develop effective treatments for patients by tailoring to their individual demographic, clinical, and/or genetic characteristics. To achieve this goal, statistical models must be developed that can identify and evaluate potentially heterogeneous treatment effects in a robust manner. The oft-cited existing methods for assessing treatment effect heterogeneity are based upon parametric models with interactions or conditioning on covariate values, the performance of which are sensitive to the omission of important covariates and/or the choice of their values. We propose a new Bayesian nonparametric (BNP) method for estimating heterogeneous causal effects in studies with zero-inflated outcome data which arise commonly in health-related studies. We employ the enriched Dirichlet process (EDP) mixture in our BNP approach, establishing a connection between an outcome DP mixture and a covariate DP mixture. This enables us to estimate posterior distributions concurrently, facilitating flexible inference regarding individual causal effects. We show in a set of simulation studies that the proposed method outperforms two other BNP methods in terms of bias and mean squared error (MSE) of the conditional average treatment effect estimates. In particular, the proposed model has the advantage of appropriately reflecting uncertainty in regions where the overlap condition is violated compared to other competing models. We apply the proposed method to a study of the relationship between heart radiation dose parameters and the blood level of high-sensitivity cardiac troponin T (hs-cTnT) to examine if the effect of a high mean heart radiation dose on hs-cTnT varies by baseline characteristics.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

SUPPORTING INFORMATION

The details of the posterior computation steps and the results for the imputed outcomes are available in the Supplementary Material. Software for running simulations in the two scenarios in the form of R code and complete documentation are available at <https://github.com/lit777/EDPcausal>.

Keywords

Enriched Dirichlet process; high-sensitivity cardiac troponin T; heterogeneous effects; missing at random

1 | INTRODUCTION

One goal of precision medicine is to develop effective treatments for patients by tailoring to their individual demographic, clinical, and/or genetic characteristics. Achieving this goal requires development of statistical models that are able to identify and evaluate potentially heterogeneous treatment effects in a robust manner. Models with interactions or conditioning on covariate values have dominated the literature of causal effect heterogeneity^{1,2,3,4,5}. The most popular model is a regression-based model^{1,6}. The appeal of regression models for causal inference problems in the setting of effect heterogeneity is the ease of evaluating potential changes of effects attributable to the chosen covariates by plugging in their selected values. Although this method has high efficiency in that it provides effect estimates for many different attributes/covariates at a relatively low computational cost, its use in causal analysis requires care.

One important risk is the omission of covariates inducing effect heterogeneity. This risk arises when a large number of covariates are available and analyses are restricted to a subset of these covariates due to the infeasibility of evaluating all possible relevant attributes. Even if one is able to evaluate the causal effect for given covariate values one set at a time, these models generally fail to capture the complex interplay of the numerous covariates, including interactions among covariates. There is also the additional risk of misspecification of the parametric forms of the covariates. This complexity may also be driven by the case in which the covariates have many possible discrete or continuous values, making evaluation of all possible combinations of values next to impossible. Another approach uses tree-based methods. Su et al. (2008; 2009)^{7,8} recursively partition the covariate space into two groups that have the largest difference in treatment effect. Foster et al. (2011)⁹ and Lipkovich et al. (2011)¹⁰ search for areas of covariate space where subjects display an enhanced treatment effect relative to the general population. Xu et al. (2015)¹¹ partition the covariate space based on a target function that directly relates to covariate-treatment interaction. However, tree-based methods are known to be unstable even with small changes in the data¹². Moreover, in the case of tree-based methods, there is a drawback in accurately estimating the effects and reflecting the uncertainty level of the estimated effects when the overlap condition is violated for specific covariates¹³. The overlap condition, sometimes called the common support or positivity assumption, guarantees that there is enough similarity in the distribution of potential confounders or covariates between the treated and untreated groups. In this era of big data, we anticipate that these limitations will pose a noticeable challenge in evaluating causal effect heterogeneity.

Parametric models are a popular approach to estimating causal effect heterogeneity in general. This popularity is due in part to the fact that the effects can be easily estimated by adding covariates of interest to the regression models. These conventional methods

have some limitations, however, including dependence on strong parametric assumptions. Semiparametric models such as models with splines^{14,15} are a promising alternative that does not require exact specification of the higher-order terms. However, this type of models can still be problematic due to the curse of dimensionality in the presence of numerous interactions¹⁶. Therefore, alternative estimation procedures are desirable. The emergence of Bayesian non-parametric (BNP) methods creates an opportunity to address the above limitations of the parametric model-based methods. While BNP approach is an extremely promising area of research, so far only limited BNP causal inference methods have been developed even for inference on the overall (as opposed to individual) causal effects^{17,18,19,20,21}. Hahn et al. (2020)²² proposed a variation of the Bayesian additive regression tree method, a type of BNP method, to estimate heterogeneous treatment effects, but they focus on the conditional average treatment effects (CATEs) for continuous outcome data. In contrast, our focus lies in accurately estimating individual causal effects in the presence of zero-inflated outcomes, as commonly observed in health studies. Furthermore, we aim to trace back individuals with significant effect estimates (positive or negative) and their characteristics based on the uncertainty level of each individual causal effect estimate. While tree-based methods, such as Hahn et al. (2020)²², demonstrate precise individual effect point estimates, our method serves as an alternative by properly capturing the uncertainty level of each estimate.

Zero-inflated outcome data commonly arise in health-related studies. The oft-cited existing methods in causal inference cannot adequately estimate treatment effects in the presence of excessive zeros in the outcome. Recently, some methods^{23,24} have been proposed to address the issue of zero inflation in mediation analysis, but these methods use parametric models that cannot capture complex relationships in the data, especially in the presence of high-dimensional covariates. Moreover, the main focus of these existing methods is on the estimation of average treatment effects, yet the individual baseline characteristics may be heterogeneous, thus inducing varying causal effects.

In this paper, we propose a new approach to estimating heterogeneous causal effects for zero-inflated outcome data using an enriched Dirichlet process (EDP) mixture model^{25,26,27}, a recently proposed BNP method. Oganisian et al. (2021)²⁸ proposed a similar BNP model for causal effect estimation for zero-inflated outcome data using a Dirichlet process (DP). In contrast, our BNP approach to causal effect heterogeneity has the advantage of making relatively weak modeling assumptions while having a stable algorithm (relative to tree-based methods) for parameter estimation with the use of the EDP model^{25,26}. Our BNP approach links each cluster of the outcome model to clustering models for covariates and jointly estimates posterior distributions of the parameters underlying the outcome and covariates to permit accurate inference on average treatment effects conditional on covariate values of study subjects. Based on the estimated CATEs and their associated uncertainty levels, we can track which groups have significant effect sizes (either negative or positive) and delineate their baseline characteristics.

The proposed general framework has several advantages. Firstly, as mentioned earlier, in non-overlap regions, tree-based models are unable to accurately estimate effects and reflect the corresponding uncertainty level. However, the proposed model appropriately accounts

for the uncertainty level as reflected by the observed data. Secondly, and more importantly, our approach flexibly estimates heterogeneous treatment effects that are induced by different baseline attributes (pre-treatment heterogeneity) across the population. Compared to Oganisian et al. (2021)²⁸, which uses DP for clustering the outcome and covariates jointly, our model uses EDP to allow independent clustering in the outcome model and conditional clustering in the covariate model that permits a nested clustering structure. As a result, in our proposed approach, fitting the outcome model is not heavily affected by modeling the large number of covariates, and finding subgroups based on covariates is not highly dependent on a potentially complex structure of the outcome model. Finally, a data augmentation-based imputation algorithm can be straightforwardly incorporated into the proposed model when applied to our data example where missing data are present. Specifically, this approach can address the missing covariate issue assuming a missing-at-random (MAR) mechanism.

Our proposed methodology is motivated by a study of the association between heart radiation dose parameters and the high-sensitivity cardiac troponin T (hs-cTnT) level in the blood, a candidate biomarker for early detection of chemoradiation-induced cardiotoxicity, among 225 non-small-cell lung cancer (NSCLC) patients²⁹. The chemoradiation-induced cardiotoxicity is considered related to the cardiac adverse events and the overall survival (OS) of the patients, the ultimate clinical outcome of interest. As a follow-up study to the original study, our focus is to investigate whether the effect of high mean heart radiation dose on the toxicity biomarker (hs-cTnT) may differ by individual characteristics. Findings from the current study may potentially inform personalized approaches to managing cardiotoxicity for NSCLC patients undergoing chemoradiation therapies. In practice, a value of 3.0 ng/l is the lower limit of detection of the hs-cTnT level, and often represents an appreciable portion of the outcome value. Thus, if all hs-cTnT values are shifted by -3.0 , then they can be deemed zero-inflated data.

The remainder of the article proceeds as follows. In Section 2, the target causal estimands are defined. Section 3 proposes our BNP model and the corresponding joint/conditional distributions. Section 4 presents partial details on posterior sampling, while leaving the rest of the information to the Supplementary Material. Section 5 discusses missing data imputation in the posterior computation algorithm. In Section 6, we propose a strategy for aggregating estimated CATEs based on latent clusters, which can be used for an alternative analysis separate from the target causal estimand. In Section 7, a rigorous simulation study for estimation of the heterogeneous causal effects is presented, while Section 8 illustrates the proposed approach by using the NSCLC patient data to evaluate the association between high-sensitivity cardiac troponin T and heart radiation dose parameters. The article concludes with a discussion in Section 9.

2 | CAUSAL EFFECTS

We use the potential outcomes framework³⁰ for causal inference. To simplify notation, we suppress the subject index i in this section. Denote by $Y(a)$ the potential outcome that would be observed under a treatment condition $A = a$, $a = 1$ or 0 . The average treatment effect (ATE) can be defined as a difference, $E[Y(1)] - E[Y(0)]$ (or a ratio, $E[Y(1)]/E[Y(0)]$),

as appropriate). Under the consistency assumption ³¹ (i.e., $Y(a) = Y$ when $A = a$) and the strongly ignorable treatment assignment assumption ³² (i.e., $\{Y(1), Y(0)\} \perp A | \mathbf{X}$ for a vector of confounders $\mathbf{X}$), the ATE τ can be estimated from the observed data based on the following equation

$$\tau = E[Y(1)] - E[Y(0)] = E_X[E[Y|A = 1, \mathbf{X}]] - E_X[E[Y|A = 0, \mathbf{X}]].$$

In this study, we are interested in estimating the average treatment effect conditional on specific covariate values (CATE), denoted as $\tau(\mathbf{x})$:

$$\tau(\mathbf{x}) = E[Y(1) - Y(0) | \mathbf{X} = \mathbf{x}].$$

In addition, we are interested in aggregation of the estimated CATEs associated with cohorts of subjects exhibiting varying effect sizes with each cohort sharing similar covariate values. For example, we can denote the covariate values for the group exhibiting positive $\tau(\mathbf{x})$ values as $\mathbf{X}_+$, defined as $\mathbf{X}_+ = \{\mathbf{x} \in \mathcal{X} | (1 - \alpha) \text{ C.I. } (\tau(\mathbf{x})) > 0\}$. Here, $(1 - \alpha) \text{ C.I. } (\tau(\mathbf{x}))$ denotes the $(1 - \alpha) \times 100\%$ level credible interval (C.I.) for $\tau(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}$. Similarly, the covariate values for the group displaying negative $\tau(\mathbf{x})$ values can be represented as $\mathbf{X}_-$, defined as $\mathbf{X}_- = \{\mathbf{x} \in \mathcal{X} | (1 - \alpha) \text{ C.I. } (\tau(\mathbf{x})) < 0\}$.

As this study focuses on zero-inflated outcome data, there may also be an interest in estimating the probability of observing a zero outcome value when determining effect sizes. Therefore, estimation of the following causal effect $\tau_p(\mathbf{x})$ and subsequent grouping based on $\tau_p(\mathbf{x})$ can also be conducted:

$$\tau_p(\mathbf{x}) = P[Y(1) = 0 | \mathbf{X} = \mathbf{x}] - P[Y(0) = 0 | \mathbf{X} = \mathbf{x}].$$

3 | OBSERVED DATA MODELS

Denote by $\mathbf{X}_i$ a p -dimensional vector of confounders for the i -th subject. Each covariate could be continuous or categorical. Let A_i represent a binary treatment for each subject. We assume all $\mathbf{X}_i$ precede A_i (i.e., being pretreatment covariates). Let $\mathbf{L}_i = \{A_i, \mathbf{X}_i\}$ and $\mathbf{M}_i = (1, \mathbf{L}_i)'$. For each subject i , Y_i represents a zero-inflated outcome. Then, the observed data model is specified using the EDP mixtures:

$$\begin{aligned} f(Y_i | \mathbf{L}_i, \phi_i) &= \pi_i I(Y_i = 0) + (1 - \pi_i) N(y_i | \mu_i, \sigma_i^2) I(y_i > 0) \\ L_{i,h} | \gamma_i &\sim p(l_{i,h} | \gamma_i) \quad h = 1, \dots, p+1 \\ (\phi_i, \gamma_i) | G &\sim G, \quad G \sim \text{EDP}(\alpha_\phi, \alpha_\gamma, G_0), \end{aligned} \tag{1}$$

where $\mu_i = \mathbf{M}_i' \boldsymbol{\beta}_i^\mu$, $\pi_i = f_\pi(\mathbf{L}_i; \boldsymbol{\beta}_i^\pi)$ with f_π being any function mapping $\mathbb{R}$ to $[0, 1]$, $\phi_i = \{\boldsymbol{\beta}_i^\mu, \boldsymbol{\beta}_i^\pi, \sigma_i^2\}$, for $i = 1, \dots, n$, and $G_0 = G_{0,\phi} \times G_{0,\gamma|\phi}$. Note that the probability of extra zeros, $\{\pi_i\}$, depends on the treatment and covariates $\mathbf{L}_i = \{A_i, \mathbf{X}_i\} = \{L_{i,1}, L_{i,2}, \dots, L_{i,p+1}\}$. By definition, $G = G_\phi \times G_{\gamma|\phi}$, where $G_\phi \sim \text{DP}(\alpha_\phi, G_{0,\phi})$, $G_{\gamma|\phi} \sim \text{DP}(\alpha_\gamma, G_{0,\gamma|\phi})$. From this

specification, $L_{i,h}$'s are specified to be independent within clusters, but the induced joint distribution of L_i can have a complex dependent structure (including the $A - X$ relationship). This observed data model based on the EDP mixtures specifies a model for $Y_i | L_i$ and then a model for L_i through nesting of the random partition of the L model (i.e., X and A models) within the Y model. Thus, it allows for more clusters for the X model (hereafter γ -clusters) than clusters for the Y model (hereafter ϕ -clusters), which is desirable when the dimension of X is much higher than that of Y ²⁷. The advantages of the EDP mixture model can be easily understood by comparing it with a standard DP mixture model²⁷ below for the observed data:

$$\begin{aligned} f(Y_i | L_i, \phi_i) &= \pi_i I(y_i = 0) + (1 - \pi_i) N(y_i | \mu_i, \sigma_i^2) I(y_i > 0) \\ L_i | \gamma_i &\sim p(L_i | \gamma_i) \\ \omega_i | G &\sim G, \quad G \sim DP(\alpha_\omega, G_0), \end{aligned} \quad (2)$$

where $\omega_i = (\phi_i, \gamma_i)$, $L_i = \{A_i, X_i\}$ and a DP prior for G with a concentration parameter α_ω and base distribution G_0 . When using Equation (Eq.) (2), estimating covariance matrices becomes computationally expensive as the dimension of the joint distribution of treatment and covariates increases. However, in the case of Eq. (1), the joint distribution can be easily obtained as a product of each marginal distribution, and since there is a conditional independence between covariates that are valid only within each cluster, there is no problem in identifying the final dependence structure. Furthermore, as mentioned earlier, in the case of DP mixtures that perform clustering on ω , it becomes challenging to balance the clustering induced in the outcome model and the clustering induced in the covariates as the dimension of covariates increases. However, in the case of EDP, after clustering on ϕ , nested clustering is performed on γ , which helps address this challenge.

The joint distribution of Y, A, X in Eq. (1) can be represented with the following square-breaking formulation^{25,27} (as an extension of the stick-breaking representation of the DP)³³.

$$f(Y_i, A_i, X_i) = \sum_{c=1}^{\infty} \omega_c K(Y_i | A_i, X_i, \phi_c) \sum_{j=1}^{\infty} \omega_{j|c} K(A_i, X_i | \gamma_{j|c}),$$

where the $K()$'s are the kernels of the corresponding distributions. Here, $\omega_{j|c} = \omega'_{j|c} \prod_{m < j} (1 - \omega'_{m|c})$ and $\omega_c = \omega'_c \prod_{r < c} (1 - \omega'_r)$, where $\omega'_{j|c} \sim \text{Beta}(1, \alpha_\gamma)$ and $\omega'_c \sim \text{Beta}(1, \alpha_\phi)$. The conditional distributions induced by this joint distribution are

$$\begin{aligned} f(Y_i = y | A_i = a, X_i = x) &= \sum_{c=1}^{\infty} \omega_c \left(a, x \right) K(Y_i = y | A_i = a, X_i = x, \phi_c), \\ f(A_i = a | X_i = x) &= \sum_{c=1}^{\infty} \omega_c \sum_{j=1}^{\infty} \omega_{j|c}(x) K(A_i = a | X_i = x, \gamma_{j|c}), \end{aligned}$$

where

$$\omega_c(a, x) = \frac{\omega_c \sum_{h=1}^{\infty} \omega_{h|c} K(A_i = a, X_i = x | \gamma_{h|c})}{\sum_{s=1}^{\infty} \omega_s \sum_{o=1}^{\infty} \omega_{o|s} K(A_i = a, X_i = x | \gamma_{o|s})},$$

$$\omega_{j|c}(x) = \frac{\omega_{j|c} K(X_i = x | \gamma_{j|c})}{\sum_{s=1}^{\infty} \omega_s \sum_{o=1}^{\infty} \omega_{o|s} K(X_i = x | \gamma_{o|s})}.$$

Note that the weights $\omega_c(x, a)$ and $\omega_{j|c}(x)$ are each a function of x and a , and x , respectively. Thus, the resulting conditional distribution of Y has the ability to explain complex relationships between X and A in modeling Y even though parametric kernels $K()$ are employed to fit each observation.

This model specification has several advantages over other zero-inflated data models. Compared to other parametric causal models^{23,24}, our model explicitly specifies the probability of extra zeros, π_i 's, as a nonparametric function of the treatment and covariates. Therefore, the effect of the treatment on extra zeros, as part of the treatment effect of potential interest, can be estimated. In comparison with other nonparametric models²⁸, our model is effective in the presence of high-dimensional X by allowing more γ -clusters than ϕ -clusters through enrichment of the usual DP²⁷. Specifically, if X is high-dimensional, the model in Oganisian et al. (2021)²⁸ may not represent the complex structure of the outcome model $(Y|A, X)$ adequately, due to the fact that the same DP structure is specified for the joint distribution of (Y, A, X) . Similarly, if the complexity of the outcome model dominates the DP structure, the model may not find proper clusters of the baseline covariates (γ -clusters). While Roy et al. (2018)²⁷ also use EDP to specify the observed data model, their approach differs from ours in that they assume the kernel of the outcome model to be an exponential family distribution. In contrast, the proposed method explicitly assumes a zero-inflated model as the kernel of the outcome distribution, thus allowing for estimation, in particular, of the probability of observing outcome 0, which is of practical interest. In the subsequent section, we describe the estimation strategy for the heterogeneous causal effects based on Eq. (1).

4 | POSTERIOR COMPUTATION

Based on Algorithm 8 in Neal (2000)³⁴ for sampling cluster memberships and the parameters associated with the distinct clusters, we draw posterior samples of all model parameters. Here, we briefly explain the sampling steps^{25,27} and relegate the remaining details to the Supplementary Material. Please refer to the data application section (Section 8.2) for the hyper-parameter settings. Denote by $C_i = (C_{i,y}, C_{i,x|y})$ the cluster membership for observation i where $C_{i,x|y}$ indicates the cluster membership driven by X_i nested within cluster membership $C_{i,y}$, which is driven by the $Y_i|A_i, X_i$ structure. The second element ($C_{i,x|y}$) of the cluster membership starts from 1 for each cluster membership $C_{i,y}$. For a sampled cluster membership $C_i = (C_{i,y} = c, C_{i,x|y} = j)$, we update parameters ϕ_c from

$$p(\phi_c | C_{1:n}, y, a, x, \phi_{-c}, \gamma) \propto g_{0,\phi}(\phi_c) \prod_{i: C_{i,y} = c} K(y_i | l_i, \phi_c),$$

and update parameters $\gamma_{j|c}$ from

$$p(\gamma_{j|c} | C_{1:n}, a, x, \phi_c, \gamma_{-j|c}) \propto g_{0,\gamma}(\gamma_{j|c}) \prod_{i: C_i = (c, j)} K(l_i | \gamma_{j|c}),$$

where $g_{0,\phi}$ and $g_{0,\gamma}$ denote the density corresponding to $G_{0,\phi}$ and $G_{0,\gamma|\phi}$ in the base measure G_0 , respectively.

Given the posterior samples of the parameters $\{\phi, \gamma, C\} = \{\phi_i, \gamma_i, C_i; i = 1, \dots, n\}$, the average treatment effect conditional on $\mathbf{X} = \mathbf{x}$ (CATE; $\tau(\mathbf{x})$ or $\tau_p(\mathbf{x})$) is computed based on the following conditional expectation/distribution:

$$E[Y_i | A_i = a, \mathbf{X}_i = \mathbf{x}, \phi, \gamma, C] = \frac{\frac{\alpha_\phi}{\alpha_\phi + n} \left\{ \int K(l_i | \gamma) dG_{0,\gamma}(\gamma) \right\} \left\{ \int E(Y | l_i, \phi) dG_{0,\phi}(\phi) \right\} + \sum_{c=1}^{c_y} \omega_c(l_i) E(Y | l_i, \phi_c)}{\frac{\alpha_\phi}{\alpha_\phi + n} \int K(l_i | \gamma) dG_{0,\gamma}(\gamma) + \sum_{c=1}^{c_y} \omega_c(l_i)} \quad (3)$$

or

$$\begin{aligned} & P[Y_i = 0 | A_i = a, \mathbf{X}_i = \mathbf{x}, \phi, \gamma, C] \\ &= \frac{\frac{\alpha_\phi}{\alpha_\phi + n} \left\{ \int K(l_i | \gamma) dG_{0,\gamma}(\gamma) \right\} \left\{ \int P(Y = 0 | l_i, \phi) dG_{0,\phi}(\phi) \right\} + \sum_{c=1}^{c_y} \omega_c(l_i) P(Y = 0 | l_i, \phi_c)}{\frac{\alpha_\phi}{\alpha_\phi + n} \int K(l_i | \gamma) dG_{0,\gamma}(\gamma) + \sum_{c=1}^{c_y} \omega_c(l_i)} \end{aligned} \quad (4)$$

where

$$\omega_c(l_i) = \frac{n_c}{\alpha_\phi + n} \left[\frac{\alpha_\gamma}{\alpha_\gamma + n_c} \left[\int K(l_i | \gamma) dG_{0,\gamma}(\gamma) \right] + \sum_{j=1}^{J_c} \frac{n_{j|c}}{\alpha_\gamma + n_c} K(l_i | \gamma_{j|c}) \right].$$

Here, n_c and $n_{j|c}$ denote the numbers of observations within cluster membership $C_{i,y} = c$ and cluster membership $C_i = (c, j)$, respectively. Also, c_y is the number of unique values of $C_{i,y}$ and J_c is the number of unique values of $C_{i,x|y}$ given $C_{i,y} = c$.

5 | MISSING COVARIATES

The presence of missing values in health data is common. Analysis based only on complete observations may lead to biased estimates and reduce power due to small sample sizes. Under the assumption of missing at random (MAR) ³⁵ as in Roy et al. (2018) ²⁷, we can also impute the values using a data augmentation algorithm ³⁶ by treating the missing values as

additional parameters. Denote by $X_{p,i}$ the p th covariate for observation i . If the $X_{p,i}$ value is missing, then we treat the value as an additional latent parameter that needs to be updated during MCMC using the following conditional:

$$p(x_{p,i} | a, x, y, \phi, \gamma, C_{1:n}) \propto K(x_{p,i} | \gamma_{j|c}) K(y_i | I_i, \phi_c),$$

if the i th subject belongs to cluster membership $C_i = (C_{i,y} = c, C_{i,x|y} = j)$. Sampling can be straightforwardly conducted using the Metropolis-Hastings algorithm. This data augmentation approach is similar in spirit to the multiple imputation approach in the frequentist framework. In the application (Section 8), we use this approach to impute missing baseline TnT values.

6 | ADDITIONAL ANALYSIS: AGGREGATION OF ESTIMATED CATES

In addition to the target causal estimands (CATE) introduced in Section 2 and the process of identifying individuals with significant effect sizes and their corresponding baseline characteristics, the proposed observed data models can potentially generate clustering results of baseline covariates as a by-product. Using the posterior samples, it becomes possible to perform clustering of baseline characteristics and to aggregate the estimated CATEs within each cluster. This clustering is based on the same principle as clustering baseline covariates without outcome Y by applying a Dirichlet process (DP) prior. In essence, extracting information from nested X clusters in EDP mixtures to derive groups sharing similar baseline characteristics yields similar outcomes to applying DP to the entire covariate space for clustering. Ultimately, it is a process that involves simultaneously estimating the groups and utilizing CATE values computed through EDP mixtures to calculate the group-wise average. Similarly, another paper proposed by Zorzetto et al. (2024)³⁷, utilizing another BNP method, also suggests reporting group-specific CATE based on X partition information among Y -model and X -model random partitions. However, it is important to note that these aggregated averages of CATEs should be interpreted as summaries of CATEs within each covariate-formed cluster, rather than as precisely defined causal effects. However, if the primary method employed in this study estimates CATEs and retrospectively identifies covariate values (and a cohort having these values) with significant effect sizes and their corresponding baseline characteristics, this additional analysis can complement the analysis by forming clusters defined by baseline characteristics, and examining the summarized effect size within each cluster to explore if similar conclusions can be drawn. It involves estimating the effect size within each cluster based on the covariate clustering derived from the same posterior sampling process. Therefore, analyzing both sets of results can lead to more comprehensive conclusions. In this section, we provide a concise overview of this approach, with further details being presented in the Supplementary Material.

For clusters $k = 1, \dots, K$ formed based on baseline covariates $\mathbf{X}$, the aggregated summary for the k -th cluster is defined as follows:

$$\tau_k = E[Y(1) - Y(0) | K = k] = E_{\mathbf{X}|K=k}[E[Y | A = 1, \mathbf{X}, K = k]] - E_{\mathbf{X}|K=k}[E[Y | A = 0, \mathbf{X}, K = k]],$$

where subjects within each cluster have similar values of X .

With posterior samples from the r th Markov chain Monte Carlo (MCMC) iteration, we draw a two-dimensional cluster membership variable $C_i^{(r)} = (C_{i,y}^{(r)}, C_{i,x|y}^{(r)})$ for subject $i = 1, \dots, n$, where $C_{i,x|y}^{(r)}$ indicates cluster membership for X_i that is nested within cluster membership $C_{i,y}^{(r)}$ for Y_i . We repeat the following steps (for $r = 1, 2, \dots, M$):

1. Find unique (vector) values in $(C_1^{(r)}, C_2^{(r)}, \dots, C_n^{(r)})$ and denote them as $\tilde{c}^{(r)} = \{(c_{1,y}^{(r)}, c_{1,x|y}^{(r)}), (c_{2,y}^{(r)}, c_{2,x|y}^{(r)}), \dots, (c_{u^{(r)},y}^{(r)}, c_{u^{(r)},x|y}^{(r)})\}$ where $u^{(r)} \leq n$.
2. Let $n^{(r)}$ denote the number of unique values in the first coordinate of $\tilde{c}^{(r)}$ (i.e., $(c_{1,y}^{(r)}, c_{2,y}^{(r)}, \dots, c_{u^{(r)},y}^{(r)})$) and $n_c^{(r)}$ denote the number of unique values of the second coordinate of the subset of $\tilde{c}^{(r)}$ where the first coordinate is set to c ($1 \leq c \leq n^{(r)}$). Set $n^{(r)} \times n$ matrix $B_0^{(r)}$ where $b_{h,i}^{(r)}$ denotes the (h, i) th element of $B_0^{(r)}$. For each $h = 1, \dots, n^{(r)}$, find all i indexes ($1 \leq i \leq n$) with $C_{i,y}^{(r)} = h$ and denote the corresponding index set by $\mathcal{J}_{0,h}^{(r)}$. Then, assign value $C_{i,y}^{(r)}$ to $b_{h,i}^{(r)}$ for each $i \in \mathcal{J}_{0,h}^{(r)}$.
3. Set $n \times n$ matrix $B_1^{(r)}$, where $b_{i,j}^{(1)}$ denotes the (i, j) th element of $B_1^{(r)}$. For each $i = 1, \dots, n$, find all j indexes with $b_{i,i}^{(r)} = b_{i,j}^{(r)}$ and denote them by $\mathcal{J}_{1,i}^{(r)}$. Assign 1 to $b_{i,j}^{(1)}$ for $j \in \mathcal{J}_{1,i}^{(r)}$ and 0 for $j \notin \mathcal{J}_{1,i}^{(r)}$. Repeat the same steps to set $B_2, B_3, \dots, B_{n^{(r)}}^{(r)}$ with $b_{h,i}^{(r)} = b_{h,j}^{(r)}$ for $h = 2, 3, \dots, n^{(r)}$.
4. Set $B^{(r)} = \frac{1}{n^{(r)}} \sum_{h=1}^{n^{(r)}} B_h^{(r)}$. This matrix is the r th posterior sample for the posterior similarity matrix³⁸.

After the MCMC iterations, we average all $B^{(r)}$'s (after a sufficient burn-in period) element-wise to obtain the posterior mean of the similarity matrix, denoted as B_{sim} . Note that since γ -clusters are nested within ϕ -clusters, the second coordinate of $C_i^{(r)}$ is meaningful only within the first coordinate of $C_i^{(r)}$. Thus, each row of matrix $B_0^{(r)}$ in Step 2 only captures clustering information of X within a certain ϕ -cluster where it belongs. However, different ϕ -clusters are explored across the MCMC iterations, which accumulate sufficient information about the overall γ -clusters. In the Supplementary Material, we provide the posterior similarity matrices from a toy example (Figure S5).

Once the posterior similarity matrix B_{sim} is obtained, we find the best partition using PAM³⁹ where centroids should be elements in the data set. This step is similar to the postprocessing procedure described in Kim et al. (2018)¹⁶. Then, in each partition ($k = 1, \dots, K$), we identify subjects that belong to the partition and summarize posterior estimates of average treatment effects conditional on covariate values of the corresponding subjects. This represents the aggregated summary of estimated CATEs for each group (or partition) $k = 1, \dots, K$.

7 | SIMULATION

We examine the performance of the proposed method (denoted as EDP+ZI) in comparison with two different methods: (I) Dirichlet process mixture of zero-inflated regression

(DPglm) ²⁸; (II) Bayesian causal forest (BCF) method ²². Our specific focus is on comparing the ability of each method to estimate effect sizes and capture the uncertainty level when the overlap assumption is violated. We examine two distinct simulation scenarios.

Specifically, for the simulated data ($n = 600$), we generate one binary covariate (X_1), one discrete covariate (X_2), and three continuous covariates ($X_3 - X_5$) as effect modifiers. The binary treatment (A) is generated using a logistic regression model, while the outcome is generated from a mixture of a point mass at 0 and a truncated normal distribution. The mixture probability π for generating a 0 outcome value is also derived from a logistic regression model. Table 1 describes the data generating mechanism for Scenarios 1 and 2.

The above data generating mechanism ensures, in Scenario 1, that the degree of violation of the overlap assumption is moderate, while in Scenario 2, that the violation of the overlap assumption is severe. Figure 1 illustrates the two distinct scenarios of non-overlap, characterized by different magnitudes, based on a single set of replicate data points. In Scenario 1, there is overlap between the two groups when considering the values of X_2 within the range of -1 to 2 . However, the overlap region becomes significantly narrower in Scenario 2. Both competing models (DPglm and BCF) employ BNP methods to provide flexible effect estimation, making them suitable for comparison. However, as they target individual treatment effects (ITEs) or average treatment effects (ATEs) rather than CATEs, direct comparisons become challenging. Therefore, for the two competing models, we categorized subjects into groups based on unique values of a discrete covariate (X_2) and compared the average ITE within these groups to the proposed model's average treatment effects conditional on each X_2 value (CATEs). In both the proposed model (EDP+ZI) and DPglm, the mean structure of the outcome model consists of a linear combination of treatment A and covariates X_1 through X_5 , without incorporating any interaction terms or non-linear terms. This can be expressed as $\mu_i = \beta_{i,0}^\mu + \beta_{i,1}^\mu A_i + \beta_{i,2}^\mu X_{i,1} + \dots + \beta_{i,p+1}^\mu X_{i,p}$ and $\log(\pi_i/(1 - \pi_i)) = \beta_{i,0}^\pi + \beta_{i,1}^\pi A_i + \beta_{i,2}^\pi X_{i,1} + \dots + \beta_{i,p+1}^\pi X_{i,p}$. Note that both interaction and non-linear terms are present in the simulation truth of Scenarios 1 and 2 (as demonstrated in Table 1). We specified a $Gamma(1, 1)$ prior for the concentration parameter (α) in the DPglm. $Gamma(1, 1)$ was also specified for the concentration parameter (α_θ) of the outcome model in the EDP. Additionally, for the concentration parameter (α_r) related to nested clustering, which is unique to the EDP model, $Gamma(1, 1)$ was also assigned. In the base measure, both the DPglm and EDP specified independent $Normal(0, 10)$ priors for the coefficients of the outcome model, a $Uniform(0, 1)$ for the success probability parameter of the binary covariate, and in the EDP, a $Normal(0, \tau^2/c_0) \times Inv. \chi^2(\nu_0, \tau_0^2)$ prior (with $c_0 = 1$, $\nu_0 = 2$, $\tau_0 = 1$) was assumed for the mean (μ) and variance (τ^2) of the continuous covariate, while the DPglm was assigned independent $Normal(0,10)$ and Inverse Gamma priors (referencing empirical variance). We specified an inverse gamma prior as the base measure for the variance parameter (σ^2) in the outcome models of both DPglm and EDP. As the main result, we presented the outcomes of specifying an $Inv. Gamma(shape = 10, rate = 10000)$ for the variance parameter of the DP and EDP models in Scenarios 1 and 2. The results of

specifying an *Inv. Gamma*($shape = 1, rate = 1$) are summarized in Scenarios S1 and S2 of the supplemental material.

Figure 2 presents the average estimated effects for each $X_2 = x$ value (i.e., $\tau(x)$ and $\tau_p(x)$) across 200 replicate datasets in Scenario 1. The true CATEs are represented by black dots, while the estimated effects from each method are depicted by red asterisks (EDP+ZI), blue X marks (BCF), and green crosses (DPglm). However, in the case of BCF, no separate estimation for $P(Y = 0|A = a)$ is performed given that BCF is applicable only to continuous outcomes, thus resulting in no outcome for $\tau_p(x)$ (right panel of Figure 2). For $\tau(x) = E[Y(1) - Y(0)|X_2 = x]$, both the EDP+ZI and BCF models provide more accurate estimation than the DPglm model. Specifically, while the BCF produced estimates closer to the true value, it cannot calculate $P(Y = 0|A = a)$ and, as will be explained in more detail below, there are issues with posterior uncertainty quantification. Therefore, the proposed method is overall preferred. Especially in scenario 2, where non-overlap becomes more severe, the proposed method is also competitive with BCF in terms of point estimation. The DPglm model, which employs a single DP structure in the joint model of the outcome and covariates, demonstrates lower flexibility than the proposed method, resulting in clear biases in the estimates of $\tau(x)$ and $\tau_p(x)$. In this scenario, given a moderate number of covariates in the outcome model, it does not seem that the clustering of DPglm would be excessively influenced by the covariates. Therefore, it is expected that the effect of more refined clustering through the nested structure of EDP is superior to relatively fewer, coarse clustering results obtained from a single DPglm model in general. In the Supplementary Material, we provide a detailed explanation along with illustrative Figures S1 and S7. In summary, in regions where non-overlapping becomes severe (especially in Scenario 2), BCF struggles to capture the true value in its credible interval. This is due to its generation of a constant-length credible interval accompanied by a biased mean estimate. In contrast, the proposed model produces relatively wider credible intervals with smaller biases in point estimates, which helps maintain coverage to some extent in these non-overlapping regions. To clarify, here, “coverage” refers to its meaning in the context of nominal frequency, and in this simulation, we prefer models that demonstrate close to nominal frequency coverage with the shortest possible credible interval width. The DPglm method also yields credible intervals which, however, are abnormally wide reflecting prior uncertainty. Based on Figure S7 (in the Supplemental Material), these results of DPglm are due to the sparse appearance of $A = 0$ observations in regions where non-overlapping becomes severe (regions with only data from $A = 1$ observations). In fact, a cluster was generated to explain these rare observations. As indicated in Eq. (2), DP clustering operates on the joint distribution of A , X , and Y . Consequently, the variance of the Y model represented by this cluster is estimated to be extremely large, as the clustering is merely based on the information from the Y model. *In contrast, the EDP+ZI model allows covariate-induced subclusters formed within the same parent cluster generated based on the outcome model, thereby allowing potential strength borrowing in estimating the CATEs in non-overlapping regions from other overlapping regions.* Based on Figure S7, during the process of clustering subjects using the outcome regression model, the rare observation does not form a cluster by itself. Therefore, the variance parameter of the outcome model sampled from that cluster reflects posterior uncertainty, which has been updated through the data rather than simply reflecting the

prior's uncertainty level. As a result, the uncertainty level for estimating the CATEs in the non-overlapping regions may potentially be brought down compared to that of the DPglm method. Table 2, for comparing the average absolute bias (abs. Bias) and mean squared error (MSE) for the average treatment effect estimates against the true values, suggests that the proposed method (EDP+ZI) outperforms the other two methods.

To gain insights into how these methods reflect the level of uncertainty, we can assess the coverage and average length of the 95% credible interval (Table 3), in addition to the point estimates of CATEs. Among the three methods, the proposed model showed the shortest credible interval while maintaining coverage close to the nominal frequency at most values (or regions) of covariate X_2 . In regions where non-overlap exists (e.g., $X_2 = 2$), the proposed model displayed satisfactory nominal coverage, with a shorter credible interval than the DP method. In contrast, the BCF method failed to maintain nominal coverage, as it produced a consistent credible interval length across all regions. This is a known limitation discussed in the literature for tree-based methods like the BART model²². This tendency becomes more pronounced in Scenario 2 (i.e., as non-overlapping intensifies, the performance of BCF in terms of coverage deteriorates). On the other hand, the DPglm method shows significantly wider credible intervals across all regions, directly reflecting the prior uncertainty level.

Figure 3 and Tables 4 and 5 summarize the results for Scenario 2. It can be observed that as the overlap region decreases, bias becomes more prominent in the estimation of CATEs (Figure 3). Once again, the proposed method demonstrates relatively superior performance in terms of estimating ATE (Table 4) compared to BCF and DPglm. In the Supplemental Material, results are presented with an *Inv. Gamma*(*shape* = 1, *rate* = 1) prior for the variance parameter of the outcome model in both the DPglm and EDP+ZI. As shown in Figures S8 and S9, the credible interval for the proposed method did not change significantly with the prior setting, whereas in the case of the DPglm method, the credible interval became considerably narrower than when an *Inv. Gamma*(*shape* = 10, *rate* = 10000) prior was used. This indicates that, in non-overlap regions, the proposed method updates posterior uncertainty based on data due to the model's structure, whereas the DPglm method directly reflects the prior uncertainty level.

We would like to point out that, in both Scenarios 1 and 2, despite that the EDP+ZI model yields smaller biases and more sensible coverage probabilities than the BCF and DPglm methods, all three methods yield relatively unsatisfactory performances in severely non-overlapping regions. This is expected as no method is anticipated to extrapolate to the non-overlap regions correctly without either correct specification of the underlying model or use of highly informative prior specification in the model.

8 | APPLICATION

8.1 | Data

We apply the proposed method to the study data that were used to evaluate the role of hs-cTnT for early detection of chemoradiation-induced cardiotoxicity and mortality in NSCLC patients, and the correlation of hs-cTnT with heart radiation dose parameters²⁹. In this study, 225 patients received a radiation total dose of at least 60 Gy and were subject to

evaluation of cardiac adverse events (CAEs). Baseline characteristics of the participating patients include age, sex, race, disease stage, tumor histology, gross tumor volume, tumor location, Karnofsky performance status (KPS) score, smoking status, pre-existing heart disease, and radiation modality (IMRT vs PSPT). For more detail about these variables, see Table 1 in Xu et al. (2021)²⁹. Along with these baseline covariates, blood samples for hs-cTnT analysis were collected before chemoradiation therapy (CRT) from 158 patients, during CRT from 181 patients and after CRT from 143 patients. Serum was separated after collection and the hs-cTnT concentrations in serum were measured. In the original study, severe CAEs (grade ≥ 3) and overall survival (OS) were two clinical endpoints where the Kaplan-Meier method was used to estimate the distribution of the time to events, and subgroup analyses were performed with linear mixed-effect regression to model the longitudinal hs-cTnT measurements.

In this paper, as a subsequent analysis of the original study²⁹, we investigate the effect of high radiation dose on the toxicity biomarker (hs-cTnT). Of particular interest is the identification of individuals with non-null effect sizes and the characterization of their corresponding baseline covariates. Here, we have two outcomes of interest: (a) maximum high-sensitivity cardiac troponin T levels (TnT-max) during CRT; and (b) final high-sensitivity cardiac troponin T levels (TnT-p). A value of 3.0 ng/L is the lowest detectable limit of the toxicity biomarker, and there are 25 (9%) and 36 (16%) patients with value 3.0 for (a) and (b), respectively. Thus, it is natural to apply our proposed method, which accounts for the zero inflation, to estimate the heterogeneous causal effect (after transforming each outcome value to have a minimum value of 0)). Specifically, to set the minimum value to 0, we use the following transformation of the two outcomes: $\log(\text{TnT-max} - 2.0)$ and $\log(\text{TnT-p} - 2.0)$. Figure 4 illustrates the distribution of each transformed outcome. We use the mean dose to the heart (heart mean dose) as the treatment and dichotomize it based on the medium value (13.5 Gy), per recommendation of the study principal investigator (PI: Dr. Zhongxing Liao). For potential effect modifiers in this analysis, we consider age, gender, pre-existing heart disease, smoking status, gross tumor volume, and baseline high-sensitivity cardiac troponin T level. These variables are also considered as confounders. Smoking status is recorded using two binary dummy variables (“have smoked” (yes/no) and “currently smoking” (yes/no)). Several variables have different numbers of missing values: TnT-max (44; 19.5%), TnT-p (82; 36.4%), heart mean dose (1; 0.4%), gross tumor volume (4; 1.7%), and baseline hs-cTnT (67; 29.7%). First, we only consider patients’ data with complete information, which yields different total sample sizes for the two outcomes: TnT-max (149) and TnT-p (118). In the subsequent analysis, we impute the missing covariate values, using the method described in Section 5, and re-analyze the data with the resulting effectively larger sample sizes. Table 6 summarizes the data.

The advantage of the proposed method is that it appropriately incorporates the uncertainty level in intervals where the overlap assumption is violated, compared to other methods. As shown in Figure 5, there is a non-overlapping interval between the two groups for a specific covariate (X_4 : gross tumor volume (GTV)). In this interval, data from the treated group are observed for larger values of X_4 (GTV), while there is no data available for the control group. In other words, a larger baseline gross tumor volume indicates a tendency to receive a

higher mean dose to the heart. Therefore, this is another reason to use the proposed method in this application.

8.2 | Prior Specification and Posterior Inference

For the (hyper-)parameters, we specify $Gamma(1, 1)$ priors (with mean = 1) on the concentration parameters α_γ and α_ϕ in the EDP that implicitly control the numbers of latent classes in the ϕ -clusters and γ -clusters, respectively. In the analysis, we specify different values of the shape and rate parameters in the $Gamma(\cdot, \cdot)$ distribution that yield similar analysis results. For the parameters in the prior distribution for continuous covariates (see Supplementary Material), we set $c_0 = 1$, $v_0 = 2$ and $\tau_0 = 1$ for the hyperparameters in the prior (that is, $X \sim N(\mu, \tau^2)$ and prior $p(\mu, \tau^2) = N(0, \tau^2/c_0) \times Inv.\chi^2(v_0, \tau_0^2)$). For the probability for each binary covariate, we assume a $Beta(0.1, 0.1)$ distribution. Each element of vectors β_i'' and β_i^* has an independent normal prior with mean 0 and variance 100. For each outcome (TnT-max or TnT-p), we run 40,000 MCMC iterations and discard the first half as burn-in. To determine the model fit, we use graphical posterior predictive checks by displaying observed data y along with predicted y from the posterior predictive distributions (results omitted).

8.3 | Results

In this application, to identify the X group (X_+) with positive effects $\tau(x)$ and/or $\tau_p(x)$, the CATEs were calculated for each subject's X value. The results are summarized in Figure 6.

For the TnT-max outcome, we found that 95 subjects had X values that exhibited a statistically significant positive CATE for $\tau(x)$, but none for $\tau_p(x)$. The estimated effect of high mean heart radiation dose on TnT-max for the positive effect group (i.e., subjects with $x \in X_+$) was 0.49, with a corresponding 95% credible interval of (0.15, 0.83). For the null effect group consisting of the remaining 54 subjects, the estimated impact of a high mean heart radiation dose on TnT-max was 0.06, with a corresponding 95% credible interval of (-0.52, 0.53). It is of further interest to track the baseline characteristics of the subjects of the positive effect group and examine how these characteristics differ from those of the remaining patients, i.e., those in the null effect group. Figure 7 presents a summary of baseline covariates for the positive effect group (Group 1; $n_1 = 95$) and the null effect group (Group 2; $n_2 = 54$). A notable difference between the two groups is that the positive effect group is composed of patients with relatively small gross tumor volume (GTV) and low TnT value before the radiation. In particular, the baseline TnT value (TnT0) stands out as a differentiating factor between the two groups. The positive effect group mainly consists of patients with the lowest baseline TnT values among the entire sample (48 patients in this group with baseline TnT values of 3.0 ng/l and 47 patients with a baseline TnT value less than 9.96 ng/l). Therefore, we may infer that significant effects of high mean heart radiation dose tend to occur in patients with very low baseline TnT values. To test for a difference between the two groups for both the GTV and TnT0 covariates, a Welch's two-sample t-test (unequal variance) was conducted. The null hypothesis of no difference in means between the two groups for each variable was rejected at a significance level of 0.01, confirming differences in these covariates between the positive and null effect groups.

For the TnT-p outcome, the estimated effects of high mean heart radiation dose on TnT-max ($\tau(\mathbf{x})$) and on the binary outcome of TnT-p being 3.0 ng/l vs. higher ($\tau_p(\mathbf{x})$) spanned from -0.12 to 0.03 for all patients, with their respective 95% credible intervals including the null effect. Consequently, we infer that there was no significant CATE observed for the TnT-p outcome, irrespective of individual characteristics.

In Section 5, we demonstrate an approach to imputing missing covariates using a data augmentation algorithm. There were 32 missing values in the baseline TnT variable (TnT0) after excluding observations with missing TnT-max outcome values, and there were 35 missing values in the baseline TnT variable after excluding observations with missing TnT-p outcome values. For the *TnT-max outcome*, the CATEs estimated for each subject's covariate values showed that $\tau(\mathbf{x})$ for 115 subjects exhibited a significantly positive effect size. Again, the group showing a positive effect had very low GTV and baseline TnT values. For the CATE on the binary outcome of being 3.0 vs. higher, none of the covariate values from subjects in the study showed significant results. For the *TnT-p outcome*, similarly the estimated effect sizes for $\tau(\mathbf{x})$ and $\tau_p(\mathbf{x})$ were very small, and their corresponding credible intervals all included zero. These results are presented in the Supplementary Material.

8.3.1 | Additional analysis on aggregation of CATEs—In Section 6, we introduced clustering based on the covariate space and proposed a method to summarize CATEs within each cluster. Without the need for additional methods, we derived results through a post-processing method as described in Section 6 using the posterior samples obtained in the main analysis. The first cluster, which comprised a larger number of subjects (99 subjects), exhibited a positive average estimate of 0.47 with a 95% credible interval of $(0.22, 0.76)$. The second cluster, consisting of 50 subjects, yielded an average estimate of 0.05 with a 95% credible interval of $(-0.20, 0.30)$. Upon analyzing the characteristics of the first cluster, it was found to primarily consist of patients with low baseline TnT values (TnT0). Therefore, we can interpret these findings as consistent with the results based on the main analysis above in Section 8.3. The corresponding results are also summarized in the Supplementary Material.

9 | DISCUSSION

We propose a Bayesian nonparametric causal model for estimating heterogeneous causal effects of treatment when the outcome of interest has excessive zeros. Our proposed method is novel in comparison to the existing methods that make strong parametric modeling assumptions. Our approach to covariate-dependent causal analysis is innovative because it overcomes some of the problems with conventional estimation of heterogeneous causal effects. Specifically, in the conventional approaches, with a large number of covariates (X), it is often infeasible to test every value of all of those covariates in order to detect a certain set of covariates and their specific values that induce causal effect heterogeneity. Those approaches also rely on strong modeling assumptions, which are vulnerable to any misspecification of the parametric forms of the covariates and their intricate interplay. Here, in addition to flexibility in the estimation of the observed data model using the EDP method, we propose an additional way to form data-driven groups based on covariate information.

The proposed method can be easily extended to data with various types of endpoints including continuous, binary, count, categorical, and/or survival outcomes. In addition to different types of outcomes, future work will include development of a Bayesian nonparametric method for estimation of heterogeneous causal mediation effects. For example, the patients' overall survival (OS) can be considered as an outcome variable, and we can investigate the extent to which mean heart radiation dose affects the outcome as mediated by TnT levels. The method development will be similar to the proposed one, yet under a more challenging setting of causal mediation. This setting includes two factors: 1) a careful specification of a mediator model in terms of the nested structure of the EDP method, and 2) a new definition of target causal estimands (heterogeneous indirect and direct effects). Also, we will develop a novel Bayesian non-parametric method for heterogeneous causal effects when data are missing not at random. In this manuscript, we assume a missing-at-random mechanism when we impute missing baseline TnT values. However, it is not implausible that these values were missing because they were either low or high. In that case, we need to account for the missing-not-at-random mechanism.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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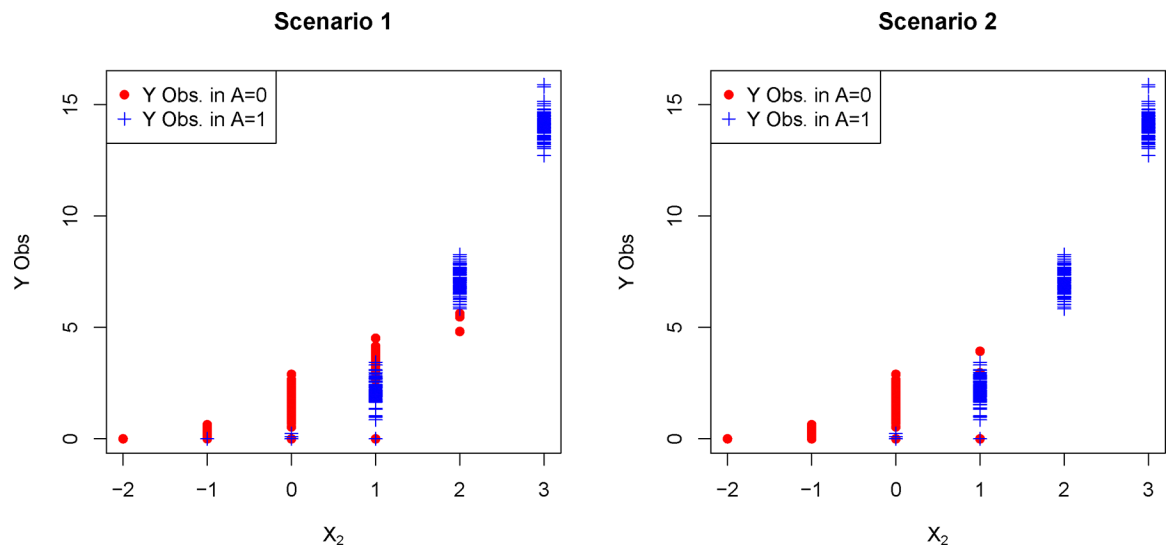
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References

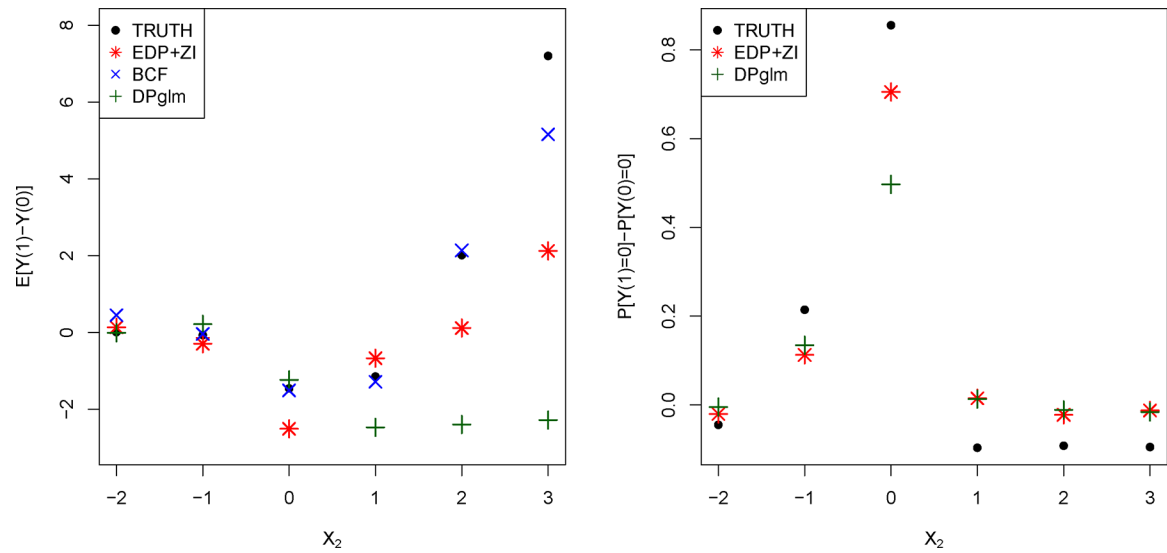
1. Robins JM. Marginal structural models versus structural nested models as tools for causal inference. In: Halloran M, Berry D., eds. *Statistical Models in Epidemiology, the Environment, and Clinical Trials*. New York, NY: Springer, 2000:95–133.
2. Angrist JD. Treatment effect heterogeneity in theory and practice. *The Economic Journal*. 2004;114(494):C52–C83.
3. Athey S, Imbens G. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*. 2016;113(27):7353–7360.
4. VanderWeele TJ. Explanation in causal inference: developments in mediation and interaction. *International Journal of Epidemiology*. 2016;45(6):1904–1908. [PubMed: 27864406]
5. Kehl V, Ulm K. Responder identification in clinical trials with censored data. *Computational Statistics & Data Analysis*. 2006;50(5):1338–1355.
6. Fairchild AJ, MacKinnon DP. A general model for testing mediation and moderation effects. *Prevention Science*. 2009;10(2):87–99. [PubMed: 19003535]
7. Su X, Zhou T, Yan X, Fan J, Yang S. Interaction trees with censored survival data. *The International Journal of Biostatistics*. 2008;4(1):1–26. [PubMed: 20231907]
8. Su X, Tsai C, Wang H, Nickerson DM, Li B. Subgroup analysis via recursive partitioning. *Journal of Machine Learning Research*. 2009;10(Feb):141–158.
9. Foster JC, Taylor JMG, Ruberg SJ. Subgroup identification from randomized clinical trial data. *Statistics in Medicine*. 2011;30(24):2867–2880. [PubMed: 21815180]

10. Lipkovich I, Dmitrienko A, Denne J, Enas G. Subgroup identification based on differential effect search - a recursive partitioning method for establishing response to treatment in patient subpopulations. *Statistics in Medicine*. 2011;30(21):2601–2621. [PubMed: 21786278]
11. Xu Y, Yu M, Zhao Y, Li Q, Wang S, Shao J. Regularized outcome weighted subgroup identification for differential treatment effects. *Biometrics*. 2015;71(3):645–653. [PubMed: 25962845]
12. Schnell PM, Tang Q, Offen WW, Carlin BP. A Bayesian credible subgroups approach to identifying patient subgroups with positive treatment effects. *Biometrics*. 2016;72(4):1026–1036. [PubMed: 27159131]
13. Zhu AY, Mitra N, Roy J. Addressing positivity violations in causal effect estimation using Gaussian process priors. *Statistics in Medicine*. 2023;42(1):33–51. [PubMed: 36336460]
14. Hastie T, Tibshirani R. Generalized additive models: some applications. *Journal of the American Statistical Association*. 1987;82(398):371–386.
15. Ruppert D, Wand MP, Carroll RJ. Semiparametric regression during 2003–2007. *Electronic journal of statistics*. 2009;3:1193. [PubMed: 20305800]
16. Kim C, Daniels M, Li Y, Milbury K, Cohen L. A Bayesian semiparametric latent variable approach to causal mediation. *Statistics in Medicine*. 2018;37(7):1149–1161. [PubMed: 29250817]
17. Hill JL. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*. 2011;20(1):217–240.
18. Karabatsos G, Walker SG. A Bayesian nonparametric causal model. *Journal of Statistical Planning and Inference*. 2012;142(4):925–934.
19. Roy J, Lum KJ, Daniels MJ. A Bayesian nonparametric approach to marginal structural models for point treatments and a continuous or survival outcome. *Biostatistics*. 2016;18(1):32–47. [PubMed: 27345532]
20. Xu Y, Müller P, Wahed AS, Thall PF. Bayesian nonparametric estimation for dynamic treatment regimes with sequential transition times. *Journal of the American Statistical Association*. 2016;111(515):921–950. [PubMed: 28018015]
21. Kim C, Daniels MJ, Marcus BH, Roy JA. A framework for Bayesian nonparametric inference for causal effects of mediation. *Biometrics*. 2017;73(2):401–409. [PubMed: 27479682]
22. Hahn PR, Murray JS, Carvalho CM. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*. 2020;15(3):965–1056.
23. Wang W, Albert JM. Estimation of mediation effects for zero-inflated regression models. *Statistics in Medicine*. 2012;31(26):3118–3132. [PubMed: 22714572]
24. Cheng J, Cheng NF, Guo Z, Gregorich S, Ismail AI, Gansky SA. Mediation analysis for count and zero-inflated count data. *Statistical Methods in Medical Research*. 2018;27(9):2756–2774. [PubMed: 28067122]
25. Wade S, Mongelluzzo S, Petrone S. An enriched conjugate prior for Bayesian nonparametric inference. *Bayesian Analysis*. 2011;6(3):359–385.
26. Wade S, Dunson DB, Petrone S, Trippa L. Improving prediction from Dirichlet process mixtures via enrichment. *The Journal of Machine Learning Research*. 2014;15(1):1041–1071.
27. Roy J, Lum KJ, Zeldow B, Dworkin JD, Re III VL, Daniels MJ. Bayesian nonparametric generative models for causal inference with missing at random covariates. *Biometrics*. 2018;74(4):1193–1202. [PubMed: 29579341]
28. Oganisian A, Mitra N, Roy JA. A Bayesian nonparametric model for zero-inflated outcomes: Prediction, clustering, and causal estimation. *Biometrics*. 2021;77(1):125–135. [PubMed: 32125699]
29. Xu T, Meng QH, Gilchrist SC, et al. Assessment of Prognostic Value of High-Sensitivity Cardiac Troponin T for Early Prediction of Chemoradiotherapy-Induced Cardiotoxicity in Non-Small Cell Lung Cancer Patients: A Secondary Analysis of a Prospective Randomized Trial. *International Journal of Radiation Oncology* Biology* Physics*. 2021;111(4):907–916. [PubMed: 34302893]
30. Rubin DB. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*. 1974;66(5):688–701.
31. VanderWeele TJ. Concerning the consistency assumption in causal inference. *Epidemiology*. 2009;20(6):880–883. [PubMed: 19829187]

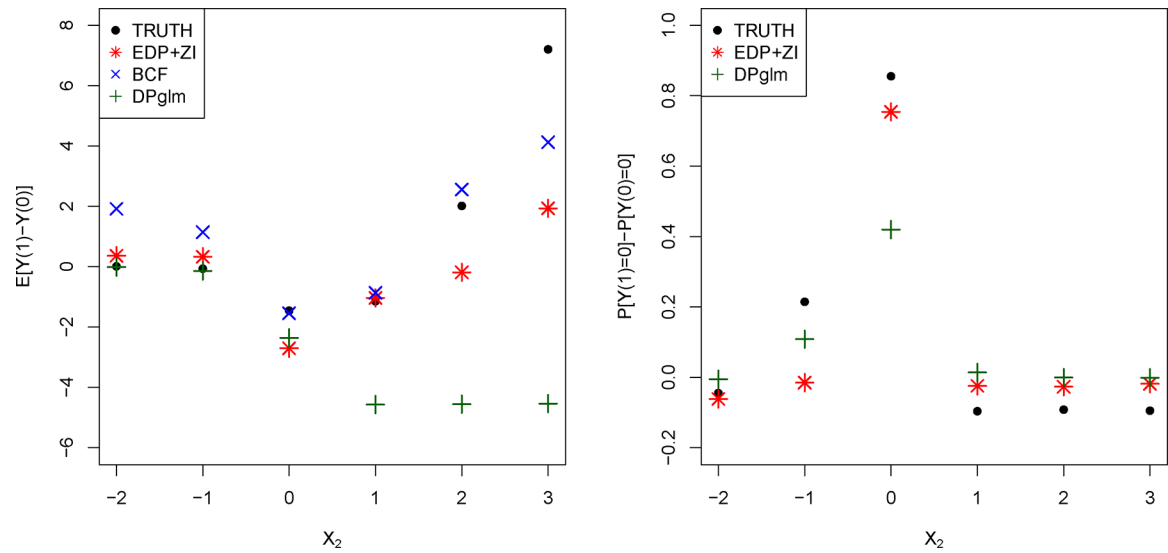
32. Rosenbaum PR, Rubin DB. The central role of the propensity score in observational studies for causal effects. *Biometrika*. 1983;70(1):41–55.
33. Sethuraman J A constructive definition of Dirichlet priors. *Statistica Sinica*. 1994;4(2):639–650.
34. Neal RM. Markov chain sampling methods for Dirichlet process mixture models. *Journal of computational and graphical statistics*. 2000;9(2):249–265.
35. Rubin DB. Inference and missing data. *Biometrika*. 1976;63(3):581–592.
36. Tanner MA, Wong WH. The calculation of posterior distributions by data augmentation. *Journal of the American Statistical Association*. 1987;82(398):528–540.
37. Zorzetto D, Bargagli-Stoffi FJ, Canale A, Dominici F. Confounder-dependent bayesian mixture model: Characterizing heterogeneity of causal effects in air pollution epidemiology. *Biometrics*. 2024 Mar 27;80(2):ujae025. [PubMed: 38640436]
38. Fritsch A, Ickstadt K, others. Improved criteria for clustering based on the posterior similarity matrix. *Bayesian Analysis*. 2009;4(2):367–391.
39. Kaufman L, Rousseeuw PJ. *Finding Groups in Data: An Introduction to Cluster Analysis*. John Wiley & Sons, 2009.

**FIGURE 1.**

The displayed data points represent the observed values for the treated and control groups in a single simulation replicate under Scenarios 1 and 2. The x -axis represents the variable X_2 . In Scenario 1 (left panel), there is a moderate non-overlap region between the observed $Y(1)$ and $Y(0)$ values based on X_2 . In Scenario 2 (right panel), there is a severe non-overlap region between the observed $Y(1)$ and $Y(0)$ values based on X_2 .

**FIGURE 2.**

[Scenario 1] The estimated conditional average treatment effects across 200 replicate datasets from each of three methods, with the true conditional average treatment effects indicated by black dots.

**FIGURE 3.**

[Scenario 2] The estimated conditional average treatment effects across 200 replicate datasets from each of three methods, with the true conditional average treatment effects indicated by black dots.

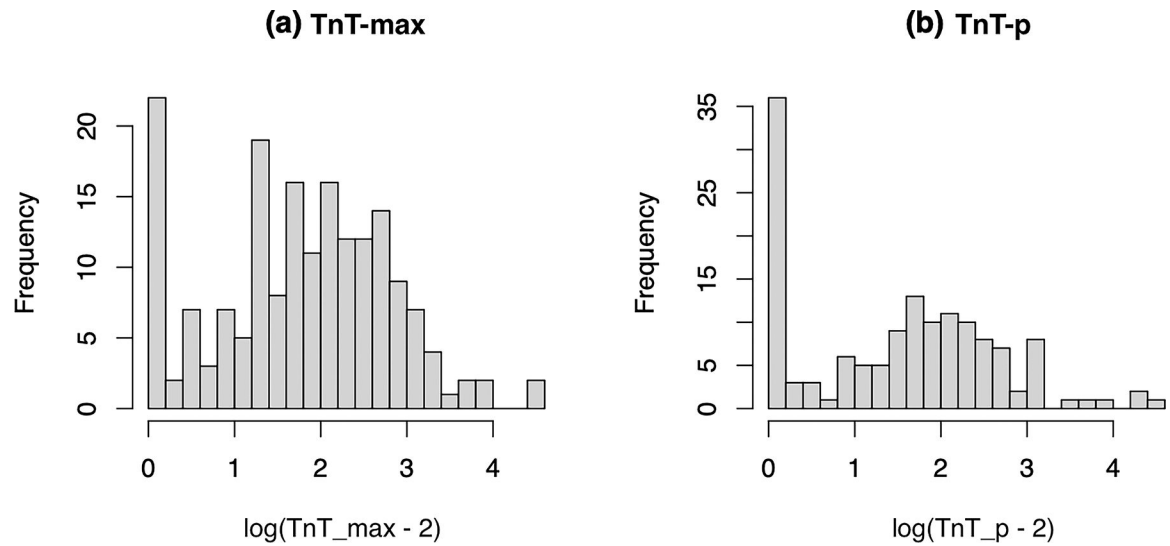
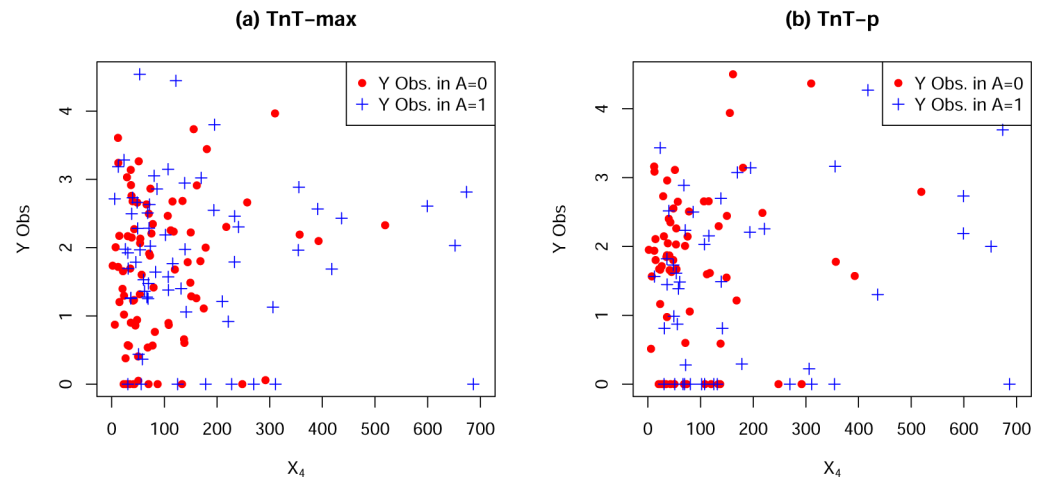
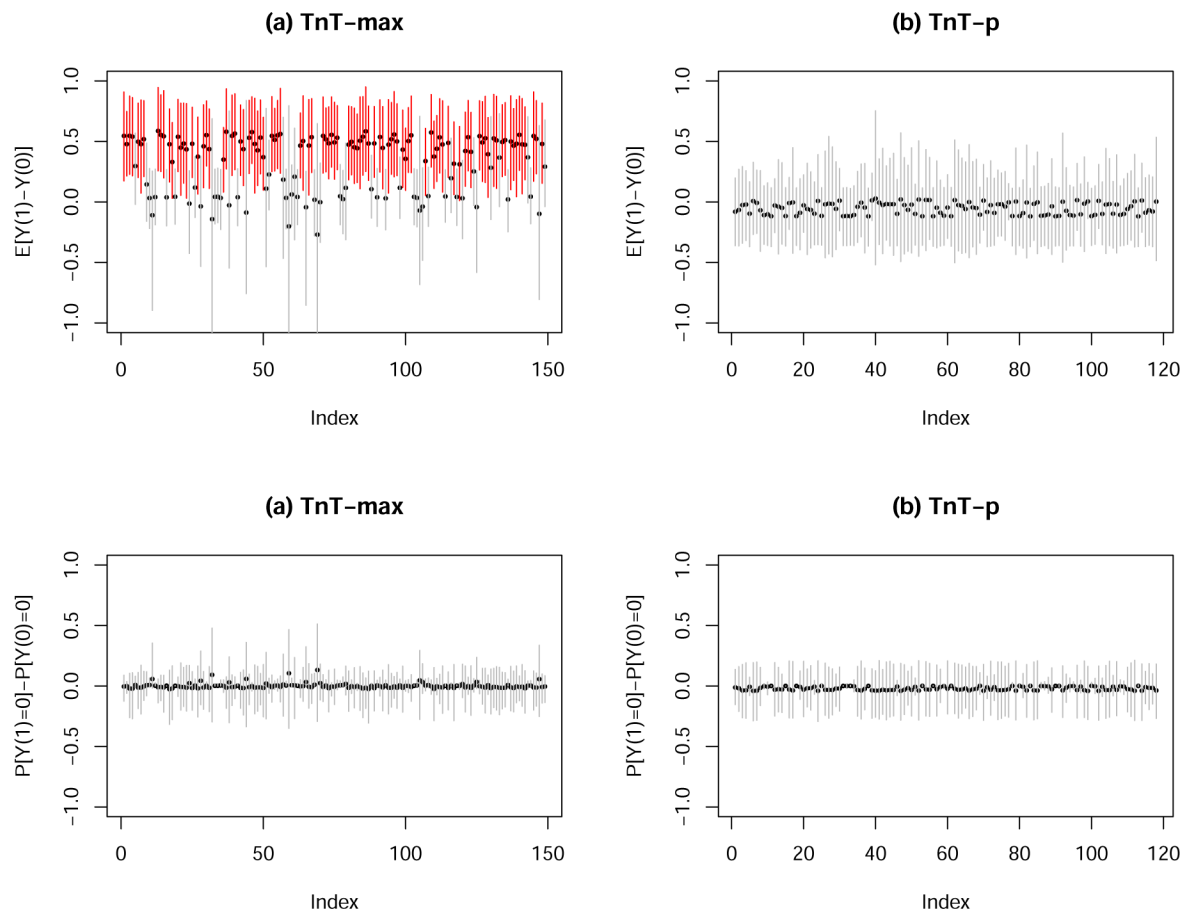


FIGURE 4.

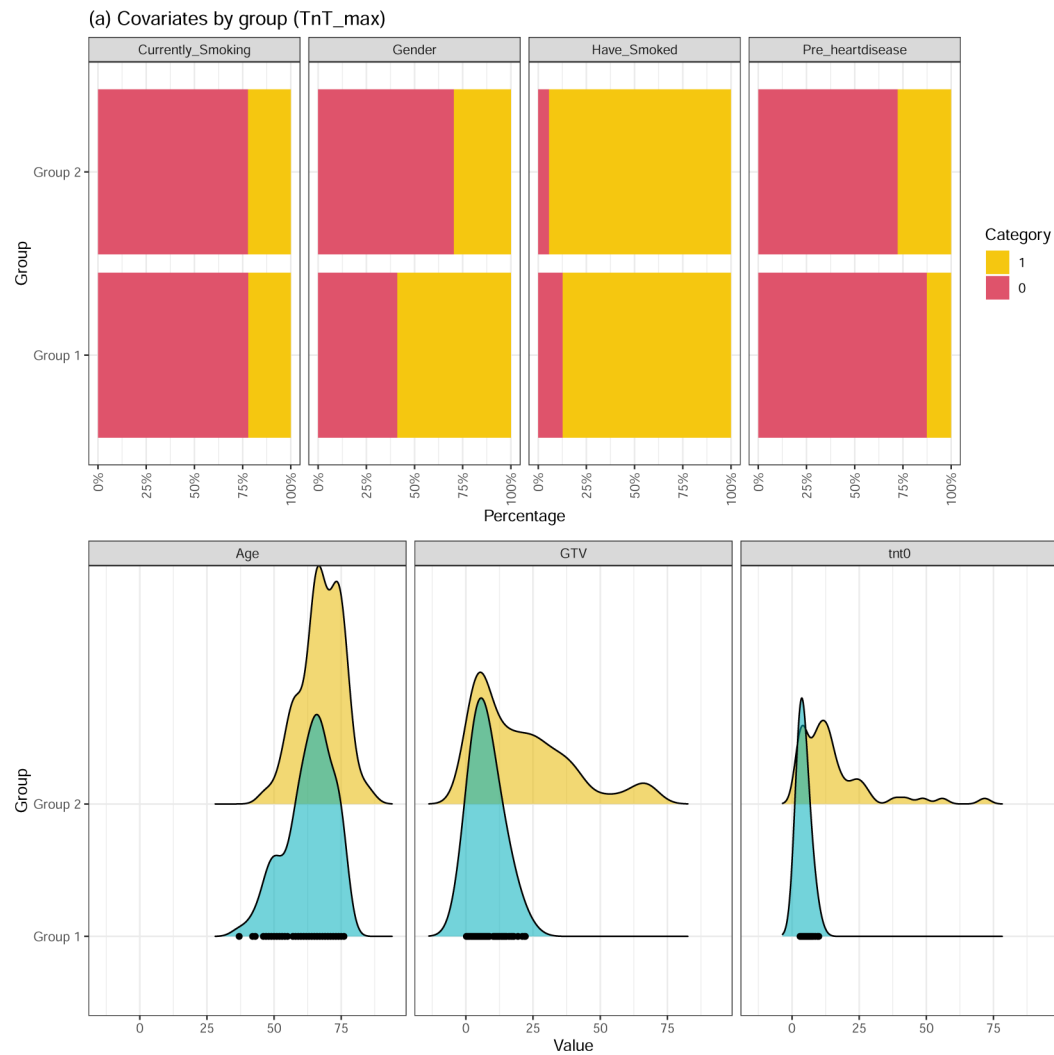
The distribution of the outcomes of interest: (a) maximum high-sensitivity cardiac troponin T levels ($\log(\text{TnT_max} - 2)$) during CRT and (b) final high-sensitivity cardiac troponin T levels ($\log(\text{TnT_p} - 2)$).

**FIGURE 5.**

The graph of the observed outcomes (Y Obs) of the two groups based on the X_4 covariate (gross tumor volume: GTV) values. In both the TnT-max and TnT-p outcomes, non-overlap between the groups occurs for larger (approximately ≥ 3) X_4 values.

**FIGURE 6.**

The posterior estimates of average treatment effects, conditioned on the actual covariate values of study subjects, provided for two outcomes: (a) the maximum high-sensitivity cardiac troponin T levels ($\log(\text{TnT-max} - 2)$) during CRT, and (b) the final high-sensitivity cardiac troponin T levels ($\log(\text{TnT-p} - 2)$). The top two figures depict $\tau(x) = E[Y(1) - Y(0) | X = x]$, while the bottom two figures illustrate $\tau_p(x) = P[Y(1) = 0 | X = x] - P[Y(0) = 0 | X = x]$. In these figures, the black dots represent the posterior mean of the effect estimates, while the gray lines depict the corresponding 95% credible intervals. The red line indicates the interval that excludes zero.

**FIGURE 7.**

[Outcome: TnT-max] The distributions of covariates in each group (Group 1: positive effect group; Group 2: null effect group). Current-Smoking: 1 (Yes) vs 0 (No); Gender: 1 (Female) vs 0 (Male); Have-Smoked: 1 (Yes) vs 0 (No); Pre-heart disease: 1 (Yes) vs 0 (No); GTV: gross tumor volume; TnT0: TnT value before radiation.

TABLE 1

Data generating process for Scenarios 1 and 2. $\text{expit}()$ denotes the inverse of the logit function. For the distribution of X_2 , $p_i = P(X_2 = i - 3) = 1/6$ for $i = 1, \dots, 6$. In Scenario 1, the β coefficient in the distribution of A is set to 2, whereas in Scenario 2, it is set to 5.

	X_1	X_2	X_3	X_4	X_5
Distribution	$\text{Ber}(0.7)$	$\{p_1, p_2, \dots, p_6\} = \{1/6, \dots, 1/6\}$	$N(0, 1)$	$N(0, 1)$	$N(0, 1)$
A	$\text{Ber}(\text{expit}(1 - X_1 + \beta X_2 - X_3 - 1 + 0.4X_4 \times X_5))$				
π	$\text{Ber}(\text{expit}(-2.5 + 0.2X_1 - 0.5X_3 - 3.5A \times I(X_2 > 0) + 0.2X_4 + 0.2X_5))$				
Y	$\pi I(Y = 0) + (1 - \pi)N(\mu, 0.1)I(Y > 0)$ where $\mu = 0.4X_1 + 2X_2 + 1A \times X_2^2 - 2.5A + 0.3 X_3 + 1 + 0.4X_4 + \exp(0.1X_5)$				

TABLE 2

[Scenario 1] The mean absolute bias (abs. Bias) and the mean squared error (MSE) for the average treatment effects calculated using three different methods.

	$E[Y(1) - Y(0)]$		$P[Y(1) = 0] - P[Y(0) = 0]$	
	abs. Bias	MSE	abs. Bias	MSE
EDP-ZI	1.27	1.85	0.02	0.001
BCF	0.56	0.48	-	-
DPglm	2.87	8.31	0.05	0.01

TABLE 3

[Scenario 1] The coverage and average length of the 95% credible intervals for individual effects at specific X_2 values ($X_2 \in \{-2, -1, 0, 1, 2, 3\}$).

			X_2 values					
			-2	-1	0	1	2	3
$E[Y(1) - Y(0)]$	EDP+ZI	Coverage	0.90	0.99	0.60	0.86	0.61	0.81
		C.I. Length	1.67	3.21	2.40	1.83	5.06	11.69
	BCF	Coverage	0.84	1	0.98	0.76	0.75	0.55
		C.I. Length	2.18	1.46	1.43	2.32	4.07	5.99
	DPglm	Coverage	0.03	1	1	1	1	1
		C.I. Length	0.95	22.17	34.30	40.94	41.29	41.61
$P[Y(1) = 0] - P[Y(0) = 0]$	EDP+ZI	Coverage	0.74	0.96	0.5	0.36	0.47	0.25
		C.I. Length	0.32	0.85	0.46	0.31	0.24	0.11
	DPglm	Coverage	0.05	0.83	0.87	0.16	0.13	0.15
		C.I. Length	0.05	1.00	1.02	0.67	0.13	0.15

TABLE 4

[Scenario 2] The mean absolute bias (abs. Bias) and the mean squared error (MSE) for the average treatment effects calculated using three different methods.

	$E[Y(1) - Y(0)]$		$P[Y(1) = 0] - P[Y(0) = 0]$	
	abs. Bias	MSE	abs. Bias	MSE
EDP-ZI	1.31	2.11	0.02	0.001
BCF	1.15	2.17	-	-
DPglm	3.79	14.39	0.07	0.007

TABLE 5

[Scenario 2] The coverage and average length of the 95% credible intervals for individual effects at specific X_2 values ($X_2 \in \{-2, -1, 0, 1, 2, 3\}$).

			X_2 values					
			-2	-1	0	1	2	3
$E[Y(1) - Y(0)]$	EDP+ZI	Coverage	0.96	1	0.43	0.95	0.94	0.95
		C.I. Length	3.99	6.61	2.59	4.22	11.17	16.45
	BCF	Coverage	0.50	0.60	0.99	0.85	0.45	0.24
		C.I. Length	4.35	3.24	1.67	1.85	3.02	3.79
	DPglm	Coverage	0.03	1	1	1	1	1
		C.I. Length	0.55	23.16	34.85	40.34	40.39	40.44
$P[Y(1) = 0] - P[Y(0) = 0]$	EDP+ZI	Coverage	0.94	1	0.82	0.66	0.72	0.45
		C.I. Length	0.67	1.41	0.44	0.35	0.29	0.19
	DPglm	Coverage	0.06	0.84	0.87	0.01	0.01	0.01
		C.I. Length	0.06	1.19	0.93	0.28	0.01	0.01

TABLE 6

Summary statistics for covariates and outcome. n_m and n_p denote the numbers of samples in the TnT-max and TnT-p outcome analysis, respectively.

	Low Heart Mean Dose (≤ 1350 cGy)		High Heart Mean Dose (> 1350 cGy)	
	$Z = 0 (n_m = 85, n_p = 71)$		$Z = 1 (n_m = 64, n_p = 47)$	
	Median / $P(X = 0)$	IQR	Median / $P(X = 0)$	IQR
Outcome Data				
max TnT value during radiation (TnT-max)	7.57	(4.38, 12.26)	9.14	(5.53, 15.66)
TnT value after radiation (TnT-p)	7.55	(3.34, 12.77)	6.41	(3.00, 11.44)
Covariate Data (TnT-max outcome analysis)				
Gender (0:Male; 1:Female)	0.53	-	0.5	-
Pre-existing heart disease (0: No; 1: Yes)	0.88	-	0.0	-
Have Smoked (0: No; 1: Yes)	0.07	-	0.14	-
Currently Smoking (0: No; 1: Yes)	0.79	-	0.77	-
Gross Tumor Volume (cm^3)	66.10	(36.24, 134.50)	104.67	(53.70, 222.92)
Age (years)	67	(62.00, 73.00)	65	(58.00, 69.25)
TnT value before radiation	4.61	(3.00, 9.96)	3.82	(3.00, 7.43)
Covariate Data (TnT-p outcome analysis)				
Gender (0:Male; 1:Female)	0.54	-	0.55	-
Pre-existing heart disease (0: No; 1: Yes)	0.78	-	0.87	-
Have Smoked (0: No; 1: Yes)	0.08	-	0.19	-
Currently Smoking (0: No; 1: Yes)	0.76	-	0.79	-
Gross Tumor Volume (cm^3)	53.50	(33.62, 118.86)	107.54	(57.13, 245.50)
Age (years)	67	(59.00, 73.50)	63	(57.00, 69.50)
TnT value before radiation (TnT0)	5.84	(3.00, 10.66)	3.84	(3.00, 7.45)